I’ll Be Back, Will You Spend?

An Emotional Perspective on Alliance Burden-Sharing

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Agendas

1. Research question
2. Theory
3. Data
4. Models
5. Results
6. Conclusion

Introduction & Research Question

- Trump, Greenland, NATO
- From 2% to 5%
- Leaders, emotions, and dynamics
- RQ: What are the effects of emotions on allies behavior?

Theoretical framework

Specific Emotions

Emotions	Effects	Reason
Anger	Negative	Diplomatic temper tantrum
Fear	Negative	Commitment issues
Joy	Positive	Encouraging, shows satisfaction
Sadness	Negative	Lack of confidence
Disgust	Negative	Display of enmity
Surprise	Negative	External shock

Transatlantic Emotions Dataset

Data

- 29 NATO allies (excluding Iceland and Sweden)
- Period: 1993 - 2023
- Panel-data structure
- N = 8,137 NATO related presidential speeches
- DV: RDef
- IV: emotions (annual average)
- Controls
  - Sentiment
  - Level of Democracy
  - Intrastate Conflicts

Methodology

Methods

- Two large language models (unsupervised)
  - For IVs: DistilBERT (a more precise version of RoBERTa) >>> various aspects in the same text
  - For sentiment: BART Large MNLI (BART) model >>> general “tone”
- Four models
  - Raw Analysis
  - The European Model (EU = 1)
  - The Democratic Model (party = “Democrats”)
  - The Republican Model (party = “Republican”)

Results & Discussion

Correlation Matrix

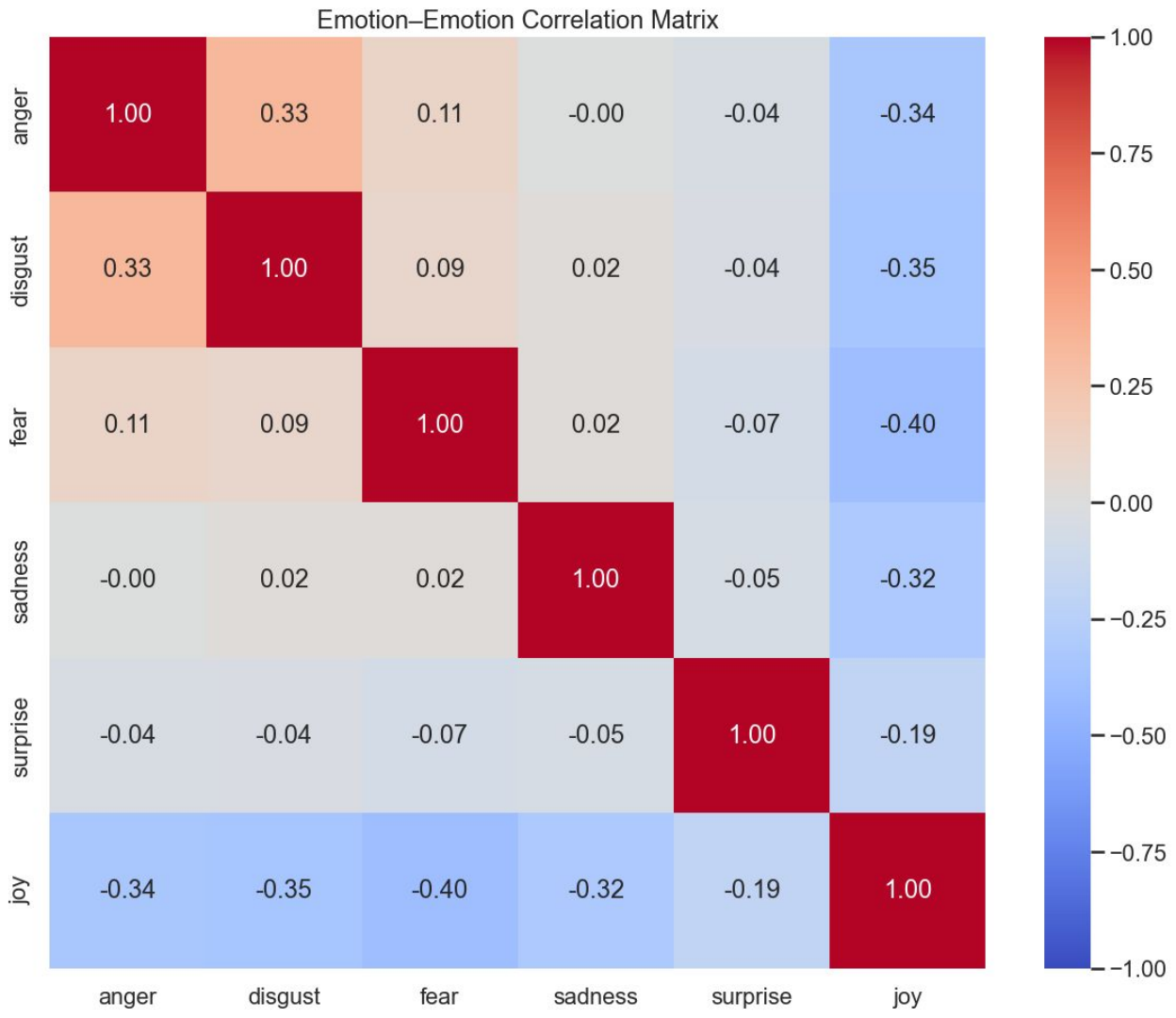

Table 2: Raw Analysis (t-1)

	Dependent variable:			
	Defense Spending (RDef)			
	Raw (1)	Country-Fixed Effects (FE) (2)	Controls (3)	Controls + FE (4)
Anger (t-1)	0.104	0.123**	0.127	0.147***
	(0.099)	(0.051)	(0.096)	(0.047)
Fear (t-1)	−0.034	−0.043**	−0.069**	−0.071***
	(0.033)	(0.017)	(0.033)	(0.016)
Joy (t-1)	−0.023	−0.018*	0.008	0.014
	(0.020)	(0.010)	(0.020)	(0.010)
Sadness (t-1)	−0.106**	−0.109***	−0.057	−0.051**
	(0.048)	(0.025)	(0.048)	(0.023)
Disgust (t-1)	−0.168	−0.169**	−0.026	−0.007
	(0.130)	(0.067)	(0.129)	(0.063)
Surprise (t-1)	−0.032	−0.042**	−0.023	−0.020
	(0.033)	(0.017)	(0.032)	(0.016)
Sentiment (t-1)			−0.039**	−0.037***
			(0.019)	(0.009)
Democracy (t-1)			−0.022***	−0.015***
			(0.004)	(0.005)
Number of Intrastate Conflicts (t-1)			0.001***	0.001***
			(0.0001)	(0.0001)

Table 3: The European Model (t-1)

	Dependent variable:			
	Defense Spending (RDef)			
	Raw (1)	Country-Fixed Effects (FE) (2)	Controls (3)	Controls + FE (4)
Anger (t-1)	0.102	0.157***	0.126	0.180***
	(0.113)	(0.057)	(0.109)	(0.052)
Fear (t-1)	−0.032	−0.038**	−0.073*	−0.074***
	(0.037)	(0.019)	(0.038)	(0.018)
Joy (t-1)	−0.025	−0.019*	0.008	0.014
	(0.022)	(0.011)	(0.022)	(0.011)
Sadness (t-1)	−0.123**	−0.125***	−0.068	−0.063**
	(0.054)	(0.027)	(0.054)	(0.026)
Disgust (t-1)	−0.185	−0.208***	−0.036	−0.042
	(0.146)	(0.074)	(0.146)	(0.069)
Surprise (t-1)	−0.036	−0.039**	−0.025	−0.014
	(0.037)	(0.019)	(0.036)	(0.017)
Sentiment (t-1)			−0.047**	−0.049***
			(0.021)	(0.010)
Democracy (t-1)			−0.025***	−0.012**
			(0.005)	(0.006)
Number of Intrastate Conflicts (t-1)			0.001***	0.001***
			(0.0001)	(0.0001)

Table 4: Democrat Model (t-1)

	Dependent variable:			
	Defense Spending (RDef)			
	Raw (1)	Country-Fixed Effects (FE) (2)	Controls (3)	Controls + FE (4)
Anger (t-1)	0.310	0.341**	0.257	0.261*
	(0.206)	(0.152)	(0.212)	(0.148)
Fear (t-1)	−0.138**	−0.131***	−0.168**	−0.127***
	(0.067)	(0.050)	(0.066)	(0.046)
Joy (t-1)	−0.080**	−0.073***	−0.022	−0.010
	(0.036)	(0.027)	(0.045)	(0.032)
Sadness (t-1)	−0.206**	−0.198***	−0.085	−0.056
	(0.081)	(0.060)	(0.074)	(0.052)
Disgust (t-1)	−0.760***	−0.766***	−0.378	−0.349**
	(0.266)	(0.197)	(0.246)	(0.173)
Surprise (t-1)	−0.145	−0.140**	−0.063	−0.062
	(0.090)	(0.066)	(0.082)	(0.057)
Sentiment (t-1)			−0.045	−0.032
			(0.048)	(0.034)
Democracy (t-1)			−0.056***	−0.098***
			(0.005)	(0.008)
Number of Intrastate Conflicts (t-1)			0.001***	0.001***
			(0.0002)	(0.0001)

Table 5: Republican Model (t-1)

	Dependent variable:			
	Defense Spending (RDef)			
	Raw (1)	Country-Fixed Effects (FE) (2)	Controls (3)	Controls + FE (4)
Anger (t-1)	0.083	0.086	0.224	0.186
	(0.163)	(0.094)	(0.179)	(0.113)
Fear (t-1)	−0.125	−0.127***	0.044	0.048
	(0.084)	(0.048)	(0.134)	(0.085)
Joy (t-1)	−0.017	−0.017	0.117	0.110**
	(0.047)	(0.027)	(0.087)	(0.055)
Sadness (t-1)	−0.072	−0.071	−0.043	−0.045
	(0.127)	(0.073)	(0.118)	(0.075)
Disgust (t-1)	0.069	0.070	0.347	0.332**
	(0.188)	(0.109)	(0.232)	(0.147)
Surprise (t-1)	−0.107	−0.109**	−0.027	−0.004
	(0.076)	(0.044)	(0.096)	(0.061)
Sentiment (t-1)			−0.075	−0.062
			(0.068)	(0.043)
Democracy (t-1)			−0.045***	−0.007
			(0.005)	(0.011)
Number of Intrastate Conflicts (t-1)			0.002*	0.002**
			(0.001)	(0.001)

Main Findings

- Anger has a positive effects on defense spending
  - potential appeasement behavior
  - Cares about reputation
- Democratic presidents are more effective than Republican presidents
- Anger, fear, and sadness are consistently significant across the models, regardless of country-fixed effects and control variables

Alternative Explanations

- Fiscal Rule Index:
  - no significance
- Proximity
  - Log transformation
  - Negative correlation
  - Significant

Table 6: FRI Model (t-1)

	Dependent variable:			
	Defense Spending (RDef)			
	Raw (1)	Country-Fixed Effects (FE) (2)	Controls (3)	Controls + FE (4)
Anger (1-year lag)	0.110	0.141**	0.130	0.163***
	(0.107)	(0.055)	(0.103)	(0.050)
Fear (1-year lag)	−0.012	−0.029	−0.055	−0.064***
	(0.036)	(0.019)	(0.037)	(0.018)
Joy (1-year lag)	−0.025	−0.019*	0.003	0.011
	(0.021)	(0.011)	(0.021)	(0.010)
Sadness (1-year lag)	−0.122**	−0.120***	−0.069	−0.064***
	(0.051)	(0.026)	(0.051)	(0.025)
Disgust (1-year lag)	−0.165	−0.180**	−0.048	−0.029
	(0.139)	(0.072)	(0.138)	(0.067)
Surprise (1-year lag)	−0.019	−0.036*	−0.015	−0.018
	(0.038)	(0.020)	(0.037)	(0.019)
Fiscal Rule Index (1-year lag)	0.00004	0.0004	0.0002	0.001*
	(0.001)	(0.001)	(0.001)	(0.0005)
Sentiment (1-year lag)			−0.048**	−0.041***
			(0.020)	(0.010)
Liberal Democracy Index (1-year lag)			−0.027***	−0.009
			(0.006)	(0.005)
Intrastate Conflict (1-year lag)			0.001***	0.001***
			(0.0001)	(0.0001)

Table 7: Proximity Model (t-1)

	Dependent variable:			
	Defense Spending (RDef)			
	Raw (1)	Country-Fixed Effects (FE) (2)	Controls (3)	Controls + FE (4)
Anger (1-year lag)	0.126	0.115	0.212	0.066
	(0.097)	(0.116)	(0.224)	(0.166)
Fear (1-year lag)	-0.075^{**}	-0.078^{**}	-0.166^{**}	0.068
	(0.034)	(0.040)	(0.070)	(0.124)
Joy (1-year lag)	0.009	0.011	−0.023	0.102
	(0.020)	(0.024)	(0.047)	(0.081)
Sadness (1-year lag)	−0.057	−0.060	−0.086	0.039
	(0.049)	(0.057)	(0.079)	(0.109)
Disgust (1-year lag)	−0.015	0.005	−0.337	0.339
	(0.129)	(0.154)	(0.261)	(0.216)
Surprise (1-year lag)	−0.021	−0.017	−0.066	0.020
	(0.033)	(0.038)	(0.087)	(0.090)
Proximity (log)	-0.005^{***}	-0.005^{***}	−0.001	-0.006^{***}
	(0.001)	(0.002)	(0.002)	(0.002)
Sentiment (1-year lag)	-0.035^{*}	-0.041^{*}	−0.051	−0.039
	(0.019)	(0.022)	(0.051)	(0.064)
Liberal Democracy Index (1-year lag)	-0.028^{***}	-0.028^{***}	-0.072^{***}	-0.042^{***}
	(0.005)	(0.006)	(0.006)	(0.006)
Intrastate Conflict (1-year lag)	0.001^{***}	0.001^{***}	0.001^{***}	0.001
	(0.0001)	(0.0002)	(0.0002)	(0.001)
Constant	0.077^{***}	0.072^{***}	0.112^{***}	0.005

Conclusion & Implications

Conclusion, Implications, Limitations

- Emotions can influence allies behavior regarding collective security.
- Some emotions do matter more: anger, fear, and sadness.
- Implications: states/leaders can securitize emotions (future research).
- Limitations
  - No pre-Cold War analysis
  - Only five presidents (3 Democrats + 2 Republicans) >>> uneven sample size
  - European = EU members, not by geographical locations

For replication, visit: https://github.com/DocMight618/us-pres-nato-allies

Q & A